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Access Without Borders: A Scoping Review to Identify Solutions to Creating Portable Identity, Education and Health Records for Refugee Children

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Abstract

Review objective: The focus of this scoping review is to identify studies, reports, and other relevant sources from the peer-reviewed and grey literature that reports on refugee children's access and barriers to portable identity, education, and health records at different stages during the migration process.

Introduction: The child refugee crisis has become a global concern as many individuals risk their lives to escape the political violence, persecution, and war. Forcibly displaced children face the genuine challenge of being unable to acquire documents pre-migration or having documents lost, stolen, or destroyed during transit. Still, little is known about refugee children's challenges when accessing and maintaining personal records while moving within and across borders. We will consolidate existing information and analyze the barriers that limit access to identity, education, and health records at different stages of migration and the policies and practices to address record portability and interoperability problems. This work is part of a program of study that seeks to understand the institutional, social, and political aspects of a refugee child's environment that potentiate their resilience under conditions of extreme stress.

Inclusion criteria: The study included academic literature focused on the production, preservation, or transmission of records pertaining to identification, education, or health of refugee children, adolescents, and young adults. The review process was limited to documents written in the English language.

Methods: We employed the framework proposed by Arksey and O'Malley [1] for the methodological synthesis of research, as well as the modifications recommended by Levac and colleagues.[2] Results of the search strategy were compiled and presented using the extended reporting guidelines for scoping reviews, as suggested by the Preferred Reporting Items for Systematic Reviews and Meta-Analyzes (PRISMA).[3] Several bibliographic and grey literature databases, including PubMed, CINAHL, Embase, Tandfonline, Proquest, Cochrane Database of Systematic Reviews, Scopus, Google Scholar, Google search engine, and UN agency websites, were searched to identify relevant studies.

Results: Using electronic literature searches, 87 articles were identified, of which 8 were deemed eligible for inclusion in this study. In addition, we identified 7 news articles and 10 reports from international organizations, most of which were focused on issues related to education record portability, as well as initiatives aimed at recognizing digital credentials for refugee young people. All eligible documents were categorized into three themes based on the analysis targets: (1) examining the challenges faced by refugees in utilizing humanitarian identity systems; (2) highlighting the need for digital credentials for refugees; and (3) exploring the potential benefits of digital health interventions for refugees. Specifically, our search showed that digital identity systems need to be more flexible and tailored to individual refugee needs. With regard to education records, the retrieved grey literature prioritizes adults' needs, but adopting digital credentials and blockchain technology may solve traditional recognition systems' constraints, ensuring that refugee children's education is also supported.

In contrast, digital health interventions, including EHRs and mobile health applications, have be better studied and shown to positively impact healthcare access, patient outcomes, and chronic disease

management for refugees in high-resource host countries. Despite this potential for positive outcomes, the portability of health records remains an under-researched and under-resourced area of a refugee child's life.

Conclusion: The increasing number of refugee children globally necessitates urgent solutions to address their need for portable identity, education, and health records. This review highlights the potential of digital interventions to improve refugee children's lives, including their resettlement outcomes. Furthermore, recent innovations like blockchain technology and digital credentials offer promising solutions for advancing equity for refugee children. Developing adaptable digital solutions and resources can enhance refugee children's access to health and education services, while also enabling them to obtain necessary documentation and identity verification for accessing basic rights and services.

Introduction

Currently, over 108 million individuals have been displaced worldwide, according to the United Nations High Commissioner for Refugees (UNHCR), of which 43% were younger than 18 years (UNHCR, 2022). The child refugee crisis has become a global concern as many individuals risk their lives to escape the instability of governments, political and community violence, persecution, and war. Here, we use the term' child refugee' to refer to a person who 'is outside their country of origin or habitual residence and is unable or unwilling to return due to a well-founded fear of persecution or harm based on their race, religion, nationality, membership of a particular social group, or political opinions' (UNHCR, no date). This may include 'individuals who face persecution or harm due to their gender identity, sexual orientation, or other similar factors' (UNHCR, no date). Child refugees may also be "displaced persons" who are defined as 'persons or groups of persons who have been forced or obliged to flee or to leave their homes or places of habitual residence, either across an international border or within a State, because of or to avoid the effects of armed conflict, situations of generalized violence, violations of human rights, or natural, or human-made disasters' (Deng and Persons, 1998. p.5).

One of the most significant but still overlooked issues that refugee children face in countries of arrival, transit, or destination is their inability to access their identity, education, and health records. Forcibly displaced children experience the very real challenge of being unable to acquire documents pre-migration or having documents lost, stolen, or destroyed during transit (Schreieck et al., 2017). The lack of documentation has been shown to compromise health, education, and resettlement outcomes, creating discontinuities in healthcare treatments, educational pathways, access to legal protections, and permanent residency (Cheesman, 2022). For instance, in the health domain, undocumented children often have no immunization records, leading to missed vaccinations and increased susceptibility to preventable diseases (Pohl et al., 2017). Similarly, the absence of a comprehensive medical history can result in misdiagnoses or inappropriate treatments, as healthcare providers lack critical information about past treatments and pre-existing conditions (Baauw et al., 2019). In education, the absence of documentation such as report cards or transfer certificates impedes accurately assessing a child's educational level (Chakrou & Keevy, 2018). Misplaced grade placements often lead to students being too

advanced or too rudimentary, which can severely hinder their educational development and adaptation to new learning environments (Chakrou & Keevy, 2018). Furthermore, the lack of documentation can make it challenging to recognize and build upon their previous educational achievements (Atesok et al., 2019; Lambrechts, 2020). From a legal perspective, undocumented refugee children often face significant barriers to accessing rights and services (Shoemaker, 2019). Without identity documents, proving their age, nationality, or family ties becomes difficult, which is crucial in asylum procedures and in accessing child protection services (Cheesman, 2022). This lack of legal recognition can lead to a precarious status, leaving them more vulnerable to exploitation and hindering their path toward stable residency or citizenship (Madon & Schoemaker,2021).

Information technology has the potential to create an interoperable system. Interoperability refers to a system's capacity to deliver timely and seamless information transfer and offers a potential means of creating portable and authenticated identification. To date, it is unclear to what extent such systems have been created for refugee children to access and transfer their identity, education, and health records.

While academic work has investigated challenges in accessing health records for refugees and migrants across all WHO regions (i.e., African Region, Region of the Americas, South-East Asia Region, European Region, Eastern Mediterranean Region, and Western Pacific Region), there is a scarcity of studies that specifically address the issue of access to identity, education, and health records among refugee children (WHO, 2012; Chiesa et al., 2019). Most available studies focus on implementing electronic health record (EHR) solutions for refugees, internally displaced persons, or asylum seekers with chronic health conditions (Buford et al., 2022). In an attempt to investigate the challenges encountered by refugee children in accessing identity, education, and health records at different stages of migration, we conducted a scoping review to identify policies and interventions that aim to meet refugee children's needs for data portability. This review examines existing best and promising practices, policies, and evidence for the effectiveness of record portability solutions relevant to refugee children and youth.

Methods

We followed Arksey and O'Malley's (2005) framework for the methodological synthesis of research, with recent adaptations suggested by Levac and colleagues (2010). A scoping review was selected over a systematic review methodology due to the limited availability of research on this subject. Studies that do exist vary greatly in their study designs and are challenging to assess for research quality. Scoping reviews offer the benefit of including all study designs, including basic descriptions of policies and interventions, without requiring a detailed evaluation of the quality of the studies that were done or the interventions that have been made. The five-step framework proposed by Arksey and O'Malley (2005) was utilized, which involved (1) identifying the research question, (2) identifying relevant studies, (3) charting the data, (4) collating selected studies and summarizing findings, and (5) reporting the results.

Research Questions

The research question (RQ) that guides this review is: What challenges do refugee children face accessing and maintaining personal records across borders? To do this, we investigated:

RQ1. How does a lack of information transfer affect refugee child and youth outcomes in cross-border contexts?

RQ2. What possible solutions/initiatives have been developed and implemented thus far for the portability, and interoperability of refugee children and youth's identity, education, and health records?

RQ3. What has been the impact of these solutions/initiatives on child and youth refugee education, health, and resettlement outcomes?

RQ4. Which of these initiatives are demonstrating effectiveness with regard to young people's developmental and social outcomes during resettlement?

Identifying Relevant Studies

Bibliographic databases were searched from 1 January 2000 to 28 February 2023: PubMed, CINAHL, Embase, Ebsco, Proquest, Cochrane Database of Systematic Reviews, Tandfonline, Google Scholar, and Scopus. Grey literature databases like UN agency websites and Google search engines were also searched. Our search strategy used Population, Concept, and Context (PCC) (*JBI Manual for Evidence Synthesis*, 2020). Considering the RQs listed above, the population includes refugees, migrant children, OR youth; the concept refers to access and barriers to identity, education, and health records; the context was the country or region of arrival, transit, or destination. Our review primarily focuses on the challenges and solutions refugee children encounter in countries of arrival, transit, and destination. We acknowledge that while the conditions in the countries of origin influence the refugee journey, our study does not delve deeply into these origins. Instead, it concentrates on the challenges faced by refugee children in maintaining or accessing vital records during and after their migration. However, it is essential to note that the detailed analysis of conditions in countries of origin is beyond the scope of this study. We used Boolean operators' AND' and 'OR' as needed to guide the search. The following keywords and their combinations were used to construct the search: refugees, migrant children, youth, adolescent, access, barriers, health records, education records, transcripts, credentials, and identity records (including specific documents such as birth certificates). As part of the subsequent phase, the bibliography of all studies identified in the previous step was examined to identify any related articles. Furthermore, the authors manually searched the reference lists of the identified articles.

Charting Data and Collation

All the articles obtained during the search were imported into Covidence, a web-based platform that streamlines screening and data extraction for systematic reviews. The review process comprised two screening stages: (1) title and abstract review and (2) full-text review. During the first stage, two reviewers independently screened the title and abstract of each retrieved citation to determine its eligibility for inclusion in the full-text review. In the second stage, both reviewers independently assessed the articles

selected for full-text review to determine whether they met the inclusion/exclusion criteria (see Table 1). In the event of any discrepancies, the full-text articles were reviewed a second time, and any disagreements were resolved through discussion until a consensus was reached. A data extraction form was developed using the RQs as a guide. Data extraction included both bibliometric details of the documents and findings relevant to the RQs. The research team progressively refined and reviewed the form to determine which variables needed to be extracted. The initial version of the data extraction form contained descriptive information about the following categories: (a) bibliometric (descriptive) information (author names; the year(s) when the study/intervention/policy was developed or conducted; title; the country(s) or region where the work was done; study objectives; method classification–qualitative, quantitative, mixed methods, review, operational reports/program evaluation; study design; study population—the number of subjects; migratory phase–transit, country of destination, arrival); (b) thematic (analytical) outcomes related to access to identity, education and health records including barriers to accessing records, and solutions/initiatives/policies to improve access to records.

Table 1
Inclusion and Exclusion Criteria

Domain	Inclusion Criteria	Exclusion Criteria
Time restriction	Literature (including grey literature) published between 1 January 2000 and 28 February 2023.	Literature published before 1 January 2000.
Language restriction	Literature available in English.	Literature published in other languages.
Population	Refugees, migrant children, youth, and adolescents[1]	Regular migrants and migrants in general
Solution	Access to Identity, education and health records. Barriers to accessing identity, education, and health records.	Access to identity, education, and health records in general, without any reference to specific solutions to making these records portable for migrants.
Setting	Country of arrival, destination[2], transit reception, displacement, WHO regions (i.e., African Region, Region of the Americas, South-East Asia Region, European Region, Eastern Mediterranean Region, and Western Pacific Region).[3]	None
Study type	Qualitative, quantitative, mixed methods study, reviews, meta-analysis, grey literature, UN agency reports, NGO reports.	None

Study Selection

During the search, 1247 articles were identified (see Fig. 1), of which 411 duplicates were removed. We reviewed all the titles and abstracts of the articles and used the pre-established criteria to exclude those irrelevant to the RQs. The remaining 87 articles were deemed eligible for inclusion if they described the experiences of refugee children or youth about access to and barriers encountered when needing identity, education, or health records at different stages of the migration process. Eight articles were chosen based on their thorough descriptions of identity and health records interventions for refugee children and youth. The initial investigation into electronic record portability across educational settings for refugee children and youth yielded limited results. Given the small number of results, an expanded search strategy was developed utilizing a range of keywords, including "migrant children," "youth," and "adolescents." This inclusion allows for a broader perspective on the challenges and solutions pertinent to portable identity records in diverse contexts. It is important to note that while our analysis includes insights relevant to migrant children, youth, and adolescents, our primary focus remains on refugee children. Our expectation is that we might identify possible solutions to record portability that have yet to be applied to refugee children and youth. This broader search yielded only one article concerned with the portability of digital records for migrants and refugees. The primary demographic focus of our review was children under the age of 18. However, recognizing the transitional challenges young people face, we included studies that extended up to 25 years of age. This inclusion acknowledges the complex realities of youth in refugee and migrant situations and aligns with our aim to explore comprehensive solutions.

While considerable academic literature discusses the portability of education records for adults seeking to transfer professional credentials, no effective digital tools are being explicitly implemented for refugee children and youth and their primary and secondary school records. In response to this, we reviewed the grey literature to identify gaps and new areas of inquiry related to the portability of education records for refugee children and youth. This phase of the scoping review yielded news articles and reports from international organizations focused on issues related to education record portability and organizations with ongoing initiatives aimed at recognizing digital credentials for young refugee people. Ultimately, eight studies were included in the review, as indicated in Appendix A. It is important to note that none of the retrieved articles or papers approach the record portability problem holistically. Each focused on only one of the three types of records child refugees require.

Our analysis of eight academic studies, seven news articles, and ten reports identified three descriptive themes that shed light on the barriers and facilitators to accessing identity, education, and health records for refugee children and youth. These themes include (1) investigating the potential benefits of digital health interventions for this population, (2) the necessity of digital credentials for child and youth refugees, and (3) examining the challenges encountered by young refugees in utilizing humanitarian identity systems.

Summarizing and Reporting Findings

Employing NVivo V.12 (QSR International), inductive thematic analysis was conducted to discover patterns and themes within the data. An initial codebook was then developed based on a sample of 8 academic articles, seven news articles, and ten reports. It used an inductive approach that allowed

patterns to emerge from the data rather than being imposed upon it. Broad categories were generated from the extracted data, focusing on evidence related to the effectiveness and experiences of digital interventions concerned with identity, education, and health records for refugee children and youth.

Results

Study Characteristics Summary:

An overview of the characteristics of the included studies is presented in Table 1. The summary highlights the diverse settings, ranging from high-income European countries to refugee camps in the Middle East. The population focus varied from children under 18 to young adults up to 25. Various study designs were incorporated, including qualitative, quantitative, meta-analysis and mixed-methods research. Our review also extended to grey literature to capture broader insights and initiatives. This included organizational reports, policy analyses, and field-specific white papers not typically subject to peer review. These documents were sourced from reputable organizations such as UNHCR and WHO, providing valuable on-the-ground perspectives and policies relevant to our research questions.

The Potential Benefits of Digital Health Interventions for Refugees

Seven of the eight selected articles reported on studies that examined the potential benefits of digital health interventions for refugee children in various contexts, including mental health screening, obesity management, infectious disease control, maternal and child health education, adverse childhood experiences identification, and vaccination support. This theme aligns with RQ1, which investigates the impact of information transfer on refugee children and youth outcomes in cross-border contexts. For example, "Baauw et al. (2019) highlight the challenges in accessing continuous medical care, which directly relates to RQ1 by demonstrating the impact of information transfer on health outcomes.

Digital health interventions, such as electronic health records (EHRs) and mobile health applications, improved access to essential health services and information, enhanced patient outcomes and facilitated the management of chronic childhood diseases. The studies also identify some limitations, such as the need for culturally specific user interfaces, addressing infrastructure factors such as internet connectivity, and improving the interoperability of health data systems. An EHR system can be a digital version of the paper records containing patients' medical and treatment histories. Such systems work optimally when all authorized health workers can access the information and make entries from multiple sites, a condition referred to in multiple papers as interoperability (Saavedra, 2013).

Baauw and colleagues (2019) investigated the logistical challenges experienced by refugee children and their families in accessing continuous medical care. The study was conducted in the Netherlands, where frequent relocation between refugee centers is a shared experience. The authors collected data from the Dutch Pediatric Surveillance Unit, which revealed that 68 cases reported by pediatricians identified many barriers to care. These barriers include limited access to medical histories (21/68), poor transfer of

medical records, ineffective communication between healthcare providers (17/68), insufficient health knowledge (17/68), and cultural differences (5/68). These factors contributed to poor quality medical care for refugee children from November 2015 to January 2017.

A considerable number of patients often required language translation that had to be provided by clinicians who were proficient in multiple languages or by professional interpreters. Storck et al. (2018) analyzed how using an EHR system helped decrease errors when transferring data from pen and paper questionnaires, mainly when working with multiple languages. They conducted their study in the context of screening for potential mental health issues among refugee minors. By using the EHR system, researchers could reduce errors in data transfer, resulting in improved accuracy and efficiency of the University Medical Center Muenster (UMCM) screening process. Notably, while this study referred to using EHR at the Center, it did not examine a better way to increase the portability of records or draw on patient records pre-migration). Additionally, the need for a culturally specific user interface to improve usability and the challenge of copying and pasting right-to-left script into left-to-right text editors resulted in wrongly allocated text.

In another study, Shapiro et al (2016) investigated weight gain among refugee children following resettlement to the Pacific Northwest or Northeast regions of the United States, finding a significant increase in Body Mass Index, with rates of overweight and obesity higher than for US children. The use of EHR helped clinicians with tracking BMI measurements. However, the study was limited in scope and intended as a pilot study to guide future research on weight gain among refugee children. Notably, the authors identified the need to obtain additional information that could provide insight into their findings, including data on the pre-resettlement environment, food insecurity, lifestyle factors, pubertal status, and mental health history, none of which was available to clinicians treating children during resettlement.

Pohl et al. (2017) examined the epidemiology and spectrum of infections in recently admitted pediatric refugees and asylum seekers in a high-income European country. Utilizing an EHR, the study found a high burden of infections caused by well-known pathogens and pre-existing non-infectious conditions in admitted patients. Most patients had an incomplete immunization status and lacked documentation of the immunizations they had received. The study had several limitations, including potential reporting bias and the exclusion of refugees and asylum seekers who were not admitted to the hospital.

A cross-sectional study by Nasir et al. (2020) highlights the successful deployment of a mobile health application by the United Nations Relief and Works Agency (UNRWA) to improve access to maternal-child health educational materials and health records for refugees registered with UNRWA's independent health system. The application, known as the e-MCH Handbook, is a newly released maternal and child health mobile application that has been instrumental in facilitating access to essential health services for vulnerable populations. The findings of this study have significant implications for the field of global health and demonstrate the potential impact of innovative digital health interventions in improving health outcomes in humanitarian settings. However, the study's limitations include the absence of iPhone users in the analysis and the need for education and reliable infrastructure to facilitate the successful use of

the e-MCH Handbook application. Guidelines for mobile Health interventions emphasize the importance of addressing infrastructure factors like internet connectivity and smartphone usage, even though very few studies explore the need for record portability. Instead, studies located for this scoping review tended to highlight some advantages to electronic record solutions but were largely focused on single place-based applications (in one country or institutional setting).

One exception was Sijilli, a cloud-based mobile EHR system from Lebanon that was created with scalability across multiple clinics and sites in mind to address the healthcare needs of Syrian refugees in low-resource settings who experience frequent migrations between camps (Shrestha et al., 2022). After each clinical visit, the system generates a password-protected, advanced encrypted standard (AES) PDF document of a patient's health record. It is stored in a key-shaped USB drive that patients can carry with them during their migration between camps. Though such a solution to data portability offers promise, the Sijilli EHR system restricts patients or providers outside of their network from modifying or updating any part of the patient's medical record, which prevents patients from having an active medical record that truly belongs to them, regardless of their location.

Syed (2022) developed clinically relevant indicators for identifying adverse childhood experiences (ACEs) using electronic health records (EHRs) of linked mothers and children in England. They found that EHRs have the potential to aid longitudinal public health monitoring and prompt early assessment of support needs for vulnerable children, including refugees. Over 70% of ACEs were identified via maternal records, recorded during primary care by General Practitioners (GPs) within two years of a child's birth, emphasizing the importance of reviewing parental and carer records to inform clinical responses to children. The study points out, however, a long-standing issue of anonymized secondary and primary care data that prevents linking children with their fathers.

Khader (2022), meanwhile, describes the Children Immunization App (CIMA), which is designed to provide mental health support and COVID-19 information to Syrian refugees in the Za'atari refugee camp. The app includes a caregiving leaflet, vaccination registration, appointment reminders, and health education. Though promising, the app's implementation has faced challenges due to limited internet connectivity and operating system incompatibility.

The Need for the Digital Credentials for Refugees

The issue of education for refugee children is a critical one that has received considerable attention in recent years. Recognizing prior qualifications is another challenge refugees face, as their qualifications may not be recognized in their host countries. This situation is particularly alarming given that education is crucial for the successful resettlement of refugee youth, as it enables them to receive recognition for the knowledge and skills that they have previously acquired that are essential for personal and professional development. This theme pertains to RQ2 and RQ3, exploring solutions for record portability and their impact on educational and health outcomes. For instance, Toker (2020) proposed a Qualifications Passport, addressing the recognition of qualifications, a critical aspect of RQ2 and RQ3.

The right to education is a fundamental human right recognized by the 1989 Convention on the Rights of the Child and the 1951 Refugee Convention. Despite this recognition, only a small percentage of refugee children have access to education while in transit (61% for primary education and 23% for secondary schooling, as reported by the UNHCR (2016)).

While the Lisbon Recognition Convention (Council of Europe, 2017) was established to facilitate the recognition of refugees' qualifications, many refugees struggle to validate their prior qualifications, as highlighted in research by Atesok and Lambrechts (Atesok et al., 2019; Lambrechts 2020).

Toker (2020) proposed a Qualifications Passport, which includes document reconstruction and an interview in the applicant's mother tongue, to address this challenge. This initiative focused on the needs of adults more than children, resulting in the Qualifications Passport piloted in 2016. Norway recommended the introduction of a European Qualifications Passport for Refugees based on this model. However, despite organizations such as World Education Services (WES) (Loo, 2016, 2017) attempting to evaluate the educational credentials of refugees who lack complete documentation, no model has met the necessary conditions of portability or interoperability.

The United Nations' Sustainable Development Goals for 2030 (SDG 4) emphasize providing inclusive and equitable education and lifelong learning opportunities, including recognizing digital credentials for refugee children. Nevertheless, the lack of an efficient "one-stop shop" universal system for collecting, storing, verifying, and connecting educational credentials across national contexts has impeded progress, as noted in a UNESCO report on digital credentialing and recognition (Chakrou & Keevy, 2018). Digital credentials represent formal and informal learning outcomes a learner achieves after fulfilling a pre-defined requirement. Only a limited number of state-funded and private collaborative initiatives, however, offer digital credentials such as CIMEA, Erasmus+, NARIC, Sony Global Education, ODEM, and IBM's Learning Credential Network (Chakrou & Keevy, 2018; Chen et al., 2018). While of potential interest for adults, none of these initiatives are tailored to the needs of children and youth and their elementary and secondary school records.

However, the development of blockchain technology may offer a possible solution in the near future. With its immutability, decentralization, security, traceability, and consensus, blockchain technology could be the basis for a robust portable credentials system that efficiently recognizes prior learning and training (Surman, 2016; Chen et al., 2018). The fundamental aim of blockchain technology is to overcome the problem of trusting strangers' data without relying on a trusted central authority to verify it. Blockchain replaces intermediaries with a permanent ledger of transactions that is distributed across a peer-to-peer network of users.

The European Blockchain Services Infrastructure (EBSI) project (2020), launched in 2019 by the European Commission in partnership with governments from member states and the European Court of Auditors (2019), is currently the most ambitious blockchain infrastructure initiative in Europe. EBSI is focused on creating cross-border government services (Ledger Insights, 2020), while initiatives like Europass 2.0 (2019), the related European Digital Credentials Infrastructure (EDCI) (2020), and eiDAS are aimed at

establishing a trusted, distributed, and shared infrastructure in the higher education ecosystem for learners to secure, own, and share their digital identity credentials. The recent initiation of these projects indicates that the adoption of blockchain technology for credentials is still nascent and has yet to gain mainstream adoption. It has also, to date, not been used to verify and move children's identity, education, or health records.

Challenges for Refugees Using Humanitarian Identity Systems

This research theme is relevant to RQ4, which focuses on the effectiveness of initiatives regarding developmental and social outcomes during resettlement. For example, Madon and Schoemaker (2021) explored digitized identities in refugee camps, linking directly to RQ4's focus on social outcomes.

Within the UN's 2030 SDGs, Target 16.9 includes mention of a digital identity that enables the establishment of inclusive societies with portable credentials. Though we could locate very few references to digital record portability for child and youth refugees, efforts in digital identity domains are shifting from theory to practice. For example, significant interest is being shown by UNHCR (UNHCR, 2018), the UN Migration Agency (2018), and global government agencies, as well as public-private initiatives being led by Microsoft, Evernym, Consensys and Accenture (Allison, 2016; ID2020, no date). The digitization of refugee identities has the potential to facilitate more personalized and customized services based on their vulnerability status. However, the lack of transparency in the decision-making processes and problems with the interoperability of multiple systems communicating with one another pose numerous risks to data governance and privacy protection, which could lead to sensitive information being compromised. The following studies shed light on refugees' challenges using humanitarian identity systems.

Madon and Schoemaker's (2021) study highlights how digitizing individual human identities, specifically refugees residing in the Bidi Bidi camp in Uganda, facilitates the UNHCR in matching them with the appropriate aid and resources. Findings suggest that document portability can provide added value through more personalized and customized services based on a refugee's changing vulnerability status. However, the interoperability of multiple systems poses a risk to data governance and protection, especially regarding sensitive issues such as child protection or domestic violence. To ensure effective data management and protection, the study underscored the importance of integrating document portability into a broader ecosystem of inter-agency arrangements, including recognizing certifications. The proposed expansion of the UNHCR's closed-loop digital identity platform to include third-party and private-sector participation will likely exacerbate these concerns.

Shoemaker (2019) examined refugees' challenges when using humanitarian identity systems, such as UNHCR's ProGres, SCOPE, and PRIMES. Using semi-structured interviews and focus groups, they found that these systems were designed for a typical refugee but limit refugees' agency and flexibility in accessing services. The lack of self-disclosure in these systems was also observed, and refugees had a limited understanding of how organizations determine their eligibility for services. The authors suggest

that greater transparency of decision-making processes could lead to a better understanding of how organizations allocate resources but could also invite "gaming" the system. The study highlights the limitations of a refugee's ability to advocate for themselves and the current design of digital identity systems that focus on the needs of organizations rather than those of individuals. To address these issues, the authors proposed the development of a participatory toolkit that would allow refugees to participate in the design or customization of UNHCR systems in their specific context: "The goal is to integrate this toolkit into PRIMES, the UNHCR's interoperable tools ecosystem" (Shoemaker et al., 2019, p. 39).

Schreieck, Wiesche, and Krcmar (2017) described the mobile application "INTEGREAT," which aims to provide refugees with relevant information upon their arrival in Germany. The app offers general and location-specific information on registration, healthcare, education, work, and daily life. The project operates as a platform that enables various information providers and stakeholders, such as municipalities, NGOs, local initiatives, and volunteers, to interact with the system. For example, municipalities provide valuable information through their offices, including social assistance. However, managing the diverse ecosystem of information providers and stakeholders presents significant challenges for INTEGRATE, including acquiring providers, identifying relevant contacts in municipalities, and managing information overflow.

Discussion

Almost all of the documents that were identified for this scoping review focused on investigating the potential benefits of digital health interventions for improving health outcomes for refugee children in high-resource host countries. The studies' strengths and relevance to our research questions lie in their focus on various healthcare areas, including mental health screening, obesity management, infectious disease control, maternal and child health education, identification of adverse childhood experiences, and vaccination support. Together, they suggest that technological solutions to enable record portability and interoperability may be possible, but this has primarily been an under-researched and under-resourced area of a refugee child's life. Our findings, however, provide evidence supporting the potential of emerging technologies like web-based applications and blockchain to successfully deploy digital health interventions that could eventually improve refugee children's lives throughout their migration journeys. At this time, these technologies have only been used to help refugees access maternal-child health educational materials and health records in a refugee child's host country (Nasir et al., 2020), facilitate the management of chronic diseases (Pohl et al., 2017), support early assessment of support needs for vulnerable children, including refugees, and aid longitudinal public health monitoring (Syed et al., 2022).

Likewise, implementing cloud-based mobile EHR systems is one of several promising approaches to record portability that could one day address the healthcare needs of refugees in low-resource settings who frequently migrate between camps (Shrestha et al., 2022). However, it is essential to ensure that patients have access to active medical records that belong to them, irrespective of their location. This is particularly important as refugees may relocate frequently, which can pose significant challenges to the

continuity of their care and effective management of chronic conditions. Therefore, ensuring that refugees have access to a secure and reliable EHR system that allows for seamless transfer of medical data across multiple sites and providers is critical to improving the quality of care for refugees (whether adults or children). This highlights the need for a patient-centric approach to healthcare that considers refugees' unique needs and challenges in low-resource settings. Moreover, logistical challenges in accessing continuous medical care for refugee children and their families have contributed to poor-quality medical care (Baauw et al., 2019). The studies we reviewed also identify several limitations to the technologies that are available, including the need for culturally specific user interfaces (Storck et al., 2018), addressing infrastructure factors such as internet connectivity (Nasir et al., 2020; Khader et al., 2022; Shrestha et al., 2022; Syed et al., 2022) and improving the interoperability of health data systems (Nasir et al., 2020).

Given that most of the digital health interventions examined in this scoping review were implemented in settings with more significant resources, more work is needed to explore the effectiveness of digital health interventions in improving health outcomes among refugee children in low-resource settings. Additionally, further research is needed to identify the specific elements of EHRs that contribute to the observed improvements in various healthcare areas in refugee settings, especially when compared to higher resource settings.

The implications for practice, policy, or research include designing culturally appropriate digital health interventions that address the unique needs of refugees and promote health equity, investing in digital health technologies to improve access to healthcare for refugees, and collaborating to overcome the challenges of addressing infrastructure factors such as internet connectivity and improving the interoperability of health data systems. Additionally, there is a need for a universally recognized international data standard or a coordinated system for collecting vital patient-encounter information to ensure that all refugees have access to high-quality healthcare.

While academic work has investigated challenges in accessing health records for refugees and migrants across all WHO regions (i.e., African Region, Region of the Americas, Southeast Asia Region, European Region, Eastern Mediterranean Region, and Western Pacific Region), (WHO, 2012; Chiesa et al., 2019), there is a scarcity of studies that specifically address the issue of access to identity records among refugee children. Only one identified study highlighted refugees' challenges using humanitarian identity systems (Madon & Schoemaker, 2021). The studies included in this review suggest that current digital identity systems are designed for a typical refugee and do not provide refugees with agency and flexibility in accessing services in ways that reflect individual needs. In discussing the concept of a 'typical' refugee, it is essential to recognize the diversity and unique challenges everyone faces. The term 'typical' here refers to everyday experiences many refugees share, such as displacement and the need for documentation. Despite this shortcoming, our review found potential solutions, including the integration of document portability into a broader ecosystem of inter-agency arrangements and the development of a participatory toolkit to allow refugees to customize digital identity systems to ensure that their unique needs and circumstances are considered (Shoemaker et al., 2019). Unfortunately, the potential benefits of

digitizing refugee identities for more personalized and customized services based on their status are offset by the risk to data governance and protection due to the lack of transparency in decision-making processes and interoperability of multiple systems. For example, UN Women's focus group research revealed that some women expressed dissatisfaction with implementing iris-scanning technology without consent (UN Women, 2020). This absence of informed consent conflicts with the principles established in the Belmont Report and is inconsistent with empowering refugees, whether adults or children (National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research, 1979). Organizations working with refugees need to consider the limitations of current digital identity systems and work to ensure that they are designed with refugees' needs in mind.

While this scoping review provides comprehensive insights into the portable identity, education, and health records for refugee children in their countries of arrival, transit, and destination, it is essential to recognize that it does not delve into the detailed analysis of conditions in the countries of origin. The circumstances in these countries often set the stage for the challenges and opportunities that arise later in the refugee journey. As such, while the findings of this review apply to the contexts addressed, they may not fully encapsulate the complexities that originate from the initial displacement. Future research could, therefore, benefit from a focused examination of the country-of-origin conditions to provide a more holistic understanding of the refugee experience.

The body of grey literature reviewed here highlights the challenges refugees face in accessing education and the recognition of their qualifications in their host countries. While almost all of the studies and reports we located that discussed the portability of education records were focused on the needs of adults, the adoption of digital credentials and blockchain technology offers some hope for a future where refugee children's education is not hindered by the limitations of traditional recognition systems that have relied on children, and their caregivers maintaining paper copies of their educational records. The reviewed studies underscore the pressing need for a standardized system to efficiently collect, store, verify, and link educational qualifications across different international settings.

Limitations

Outcomes from this scoping review ought to be interpreted cautiously. Given that this was a scoping review, we did not assess the quality of the documents we reviewed nor the studies they referred to. Nonetheless, this is consistent with the goals of scoping reviews, enabling a comprehensive evaluation of a wide range of interventions from all WHO regions (i.e., African Region, Region of the Americas, Southeast Asia Region, European Region, Eastern Mediterranean Region, and Western Pacific Region) (WHO, 2012). Moreover, this review was restricted to English and a specific period (2000–2023). This may have resulted in missed opportunities to discover innovative solutions to the portability of refugee children's health, education, and identity records. Our review's inclusive approach, considering studies on migrant youth alongside refugee children, may introduce variability in the identified contexts and challenges. While this broadens the scope of our analysis, readers should consider this aspect when interpreting our findings, especially in their direct applicability to refugee children. Despite these

constraints, our review's findings carry substantial implications for practitioners, policymakers, and donors, pinpointing promising interventions worth piloting.

Conclusion

As the number of refugee children and youth worldwide has doubled in the last decade and is projected to continue to increase, there is an urgent need for solutions to support this vulnerable population throughout their migration journey. Findings from this scoping review suggest that successfully deploying digital identity, education, and health records could improve the lives of refugee children and youth. What these documents do not show, however, is how to use technology to make refugee children's records more portable and interoperable beyond high-resource host countries where refugees resettle. Our thematic analysis has revealed critical aspects contributing to the broader discourse on the portability of refugee data. The alignment of these themes with our research questions underscores their importance in shaping future research directions and policy formulations in this domain. Given the immediacy of this problem, it is remarkable that so little attention has been paid to helping refugee children and youth create and maintain portable records. The only hopeful note we can sound is about emerging technologies that could be adapted to this problem of record continuity. Blockchain technology and digital credentials offer promising solutions for advancing social equity in healthcare and education, maintaining identity records (like birth certificates), and providing the documentation and verification necessary to enable refugee children and youth to access basic human rights protections and services.

References

1. Allison, I. (2016) *ConsenSys and Microsoft tackle human rights abuses with blockchain-based identity system*, *International Business Times UK*. Available at: httphttps://www.ibtimes.co.uk/consensys-microsoft-tackle-human-rights-abuses-blockchain-based-identity-system-1562722 s://www.ibtimes.co.uk/consensys-microsoft-tackle-human-rights-abuses-blockchain-based-identity-system-1562722 (Accessed: 19 September 2023).
2. Arksey, H. and O'Malley, L. (2005) 'Scoping studies: towards a methodological framework', *International Journal of Social Research Methodology*, 8(1), pp. 19–32. Available at: https://doi.org/10.1080/1364557032000119616.
3. Atesok, Z.O., Komsuoglu, A. and Ozer, Y.Y. (2019) 'An Evaluation of Refugees' Access To Higher Education: Case of Turkey and Istanbul University', *Journal of International and Comparative Education (JICE)*, pp. 119–136. Available at: https://doi.org/10.14425/jice.2019.8.2.119.
4. Baauw, A. *et al.* (2019) 'Health needs of refugee children identified on arrival in reception countries: a systematic review and meta-analysis', *BMJ Paediatrics Open*, 3(1), p. e000516. Available at: https://doi.org/10.1136/bmjpo-2019-000516.
5. Buford, A. *et al.* (2022) 'Systematic review of electronic health records to manage chronic conditions among displaced populations', *BMJ Open*, 12(9), p. e056987. Available at: https://doi.org/10.1136/bmjopen-2021-056987.

6. Chakrou, B. and Keevy, J. (2018) 'Digital credentialing: implications for the recognition of learning across borders - UNESCO Digital Library'. UNESCO. Available at: https://unesdoc.unesco.org/ark:/48223/pf0000264428 (Accessed: 19 September 2023).
7. Cheesman, M. (2022) 'Self-Sovereignty for Refugees? The Contested Horizons of Digital Identity', *Geopolitics*, 27(1), pp. 134–159. Available at: https://doi.org/10.1080/14650045.2020.1823836.
8. Chen, G. *et al.* (2018) 'Exploring blockchain technology and its potential applications for education', *Smart Learning Environments*, 5(1), p. 1. Available at: https://doi.org/10.1186/s40561-017-0050-x.
9. Chiesa, V. *et al.* (2019) 'Health records for migrants and refugees: A systematic review', *Health Policy*, 123(9), pp. 888–900. Available at: https://doi.org/10.1016/j.healthpol.2019.07.018.
10. Deng, F.M. and Persons, U.R. of the S.-G. on I.D. (1998) 'Report of the Representative of the Secretary-General, Francis M. Deng, submitted pursuant to Commission [on Human Rights] resolution 1997/39.' Available at: https://digitallibrary.un.org/record/251017 (Accessed: 19 September 2023).
11. European Commission (2019) *Europass Digital Credentials Infrastructure*, *FUTURIUM - European Commission*. Available at: https://ec.europa.eu/futurium/en/europass/europass-digital-credentials-infrastructure (Accessed: 19 September 2023).
12. ID2020 Alliance. (no date). Available at: https://id2020.org/alliance (Accessed: 19 September 2023).
13. IOM and APSCA (2018) *5th border management and identity conference (BMIC) on technical cooperation and capacity building*. Bangkok: BMIC. Available at: http://cb4ibm.iom.int/bmic5/assets/documents/5BMIC-Information-Brochure.pdf.
14. International Organization for Migration. (2019). Glossary on Migration. International Migration Law No. 34. Retrieved from https://www.iom.int/glossary-migration
15. *JBI Manual for Evidence Synthesis* (2020) *JBI*. Available at: https://doi.org/10.46658/JBIMES-20-01 (Accessed: 19 September 2023).
16. Khader, Y.S. *et al.* (2022) 'Scaling the Children Immunization App (CIMA) to Support Child Refugees and Parents in the Time of the COVID-19 Pandemic: A Social Capital Approach to Scale a Smartphone Application in Zaatari Camp, Jordan', *Journal of Epidemiology and Global Health*, 12(1), pp. 7–12. Available at: https://doi.org/10.1007/s44197-021-00029-x.
17. Lambrechts, A.A. (2020) 'The super-disadvantaged in higher education: barriers to access for refugee background students in England', *Higher Education*, 80(5), pp. 803–822. Available at: https://doi.org/10.1007/s10734-020-00515-4.
18. Ledger Insights. (2020) *How the EU is using blockchain to build a citizen-centric European Internet*, *Ledger Insights - blockchain for enterprise*. Available at: https://www.ledgerinsights.com/how-the-eu-is-using-blockchain-to-build-a-citizen-centric-european-internet/ (Accessed: 19 September 2023).
19. Levac, D., Colquhoun, H. and O'Brien, K.K. (2010) 'Scoping studies: advancing the methodology', *Implementation science: IS*, 5, p. 69. Available at: https://doi.org/10.1186/1748-5908-5-69.
20. Loo, B. (2016) *Recognizing Refugee Qualifications: Practical Tips for Credential Assessment*. New York, NY, USA: World Education Services (WES). Available at: https://knowledge.wes.org/wes-research-report-recognizing-refugee-credentials.html (Accessed: 19 September 2023).

21. Loo, B. (2017) *What Help Do Refugees Need to Enroll in North American Higher Education?*, *WENR*. Available at: https://wenr.wes.org/2017/05/early-insights-from-a-canada-what-help-do-refugees-need-to-enroll-in-north-american-higher-education (Accessed: 19 September 2023).
22. Madon, S. and Schoemaker, E. (2021) 'Digital identity as a platform for improving refugee management', *Information Systems Journal*, 31(6), pp. 929–953. Available at: https://doi.org/10.1111/isj.12353.
23. Nasir, S. *et al.* (2020) 'Dissemination and implementation of the e-MCHHandbook, UNRWA's newly released maternal and child health mobile application: a cross-sectional study', *BMJ Open*, 10(3), p. e034885. Available at: https://doi.org/10.1136/bmjopen-2019-034885.
24. National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research. (1979) *The Belmont Report: Ethical principles and guidelines for the protection of human subjects of research*. Available at: https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html (Accessed: 19 September 2023).
25. Pohl, C. *et al.* (2017) 'The spectrum of care for pediatric refugees and asylum seekers at a tertiary health care facility in Switzerland in 2015', *European Journal of Pediatrics*, 176(12), pp. 1681–1687. Available at: https://doi.org/10.1007/s00431-017-3014-9.
26. Saavedra, F. (2013) *Development and Evaluation of a Web-based Electronic Medical Record System Without Borders*. Thesis. Available at: https://digital.lib.washington.edu:443/researchworks/handle/1773/22424 (Accessed: 19 September 2023).
27. Schreieck, M., Wiesche, M. and Krcmar, H. (2017) 'Governing nonprofit platform ecosystems – an information platform for refugees', *Information Technology for Development*, 23(3), pp. 618–643. Available at: https://doi.org/10.1080/02681102.2017.1335280.
28. Shapiro, A. *et al.* (2016) 'Weight Trajectory in Refugee Children after Resettling in the United States: A Pilot Study', *Journal of pediatrics and child nutrition*, 2(3), p. 100115.
29. Shoemaker, E. *et al.* (2019) 'Identity at the margins: examining refugee experiences with digital identity systems in Lebanon, Jordan, and Uganda', in *Proceedings of the 2nd ACM SIGCAS Conference on Computing and Sustainable Societies*. New York, NY, USA: Association for Computing Machinery (COMPASS '19), pp. 206–217. Available at: https://doi.org/10.1145/3314344.3332486.
30. Shrestha, A. *et al.* (2022) 'Innovation is needed in creating electronic health records for humanitarian crises and displaced populations', *Frontiers in Digital Health*, 4. Available at: https://www.frontiersin.org/articles/10.3389/fdgth.2022.939168 (Accessed: 19 September 2023).
31. Storck, M. *et al.* (2018) 'Conducting a Multilingual Study Researching Traumatised Refugees Utilizing a Patient-Reported Outcome System', *Studies in Health Technology and Informatics*, 253, pp. 109–113.
32. Surman, M. (2016) *Mozilla Learning*. Available at: https://learning.mozilla.org/blog/mozillas-continued-commitment-to-open-badges (Accessed: 9 February 2023).

33. Syed, S. *et al.* (2022) ‘Identifying adverse childhood experiences with electronic health records of linked mothers and children in England: a multistage development and validation study’, *The Lancet Digital Health*, 4(7), pp. e482–e496. Available at: https://doi.org/10.1016/S2589-7500(22)00061-9.
34. Toker, H. (2020) ‘The Norwegian Way: Protection through Higher Education the Recognition Process for Syrian Refugees in HE’, *International Migration*, 58(4), pp. 101–116. Available at: https://doi.org/10.1111/imig.12664.
35. United Nations Entity for Gender Equality and the Empowerment of Women (UN Women). (2020) *Gender and the Digital Divide in Situations of Displacement: The Experiences of Syrian Refugee Women in Al-Azraq and Al-Za’atari camps*. Available at: https://jordan.unwomen.org/en/digital-library/publications/gender-and-the-digital-divide (Accessed: 19 September 2023).
36. United Nations High Commissioner for Refugees (2016) *Left Behind: Refugee Education in Crisis*, *UNHCR*. Available at: http://www.unhcr.org/59b696f44.pdf (Accessed: 19 September 2023).
37. United Nations High Commissioner for Refugees (2021) *UNHCR - UNHCR Global Trends 2022*. Available at: https://www.unhcr.org/global-trends-report-2022 (Accessed: 29 December 2023).
38. UNHCR (2018) ‘Bridging the identity divide – Is Portable User-Centric Identity Management the Answer? | UNHCR Blog’. Available at: https://www.unhcr.org/blogs/bridging-identity-divide-portable-user-centric-identity-management-answer/ (Accessed: 19 September 2023).
39. UNHCR. (2020). The UN Refugee Agency’s Handbook on Protection of Refugees. Retrieved from https://www.unhcr.org/protection/handbook/
40. UNHCR (no date) *What is a refugee?* Available at: https://www.unhcr.org/what-refugee (Accessed: 19 September 2023).
41. World Health Organization (2012) *WHO Global Observatory for eHealth. Management of patient information: Trends and challenges in Member States*, *WHO | Regional Office for Africa*. Available at: https://www.afro.who.int/publications/management-patient-information-trends-and-challenges-member-states (Accessed: 19 September 2023).
42. World Health Organization. (n.d.). WHO regional offices. Retrieved January 2, 2024, from https://www.who.int/about/who-we-are/regional-offices.

Footnotes

1. Migrants encompasses refugee children and youth, aligning with our broader search strategy (Migration, 2019). International Organization for Migration. (2019). Glossary on Migration. International Migration Law No. 34. https://www.iom.int/glossary-migration
2. 'Country of Arrival' refers to the initial country where refugees seek asylum, while 'Country of Destination' denotes the nation where refugees aim to settle. (UNHCR, 2020). UNHCR. (2020). The UN Refugee Agency’s Handbook on Protection of Refugees. https://www.unhcr.org/protection/handbook/

3. WHO regions were included to ensure a comprehensive global perspective in our review. (Organization, n.d.). World Health Organization. (n.d.). WHO regional offices. Retrieved January 2, 2024, from https://www.who.int/about/who-we-are/regional-offices.

Figures

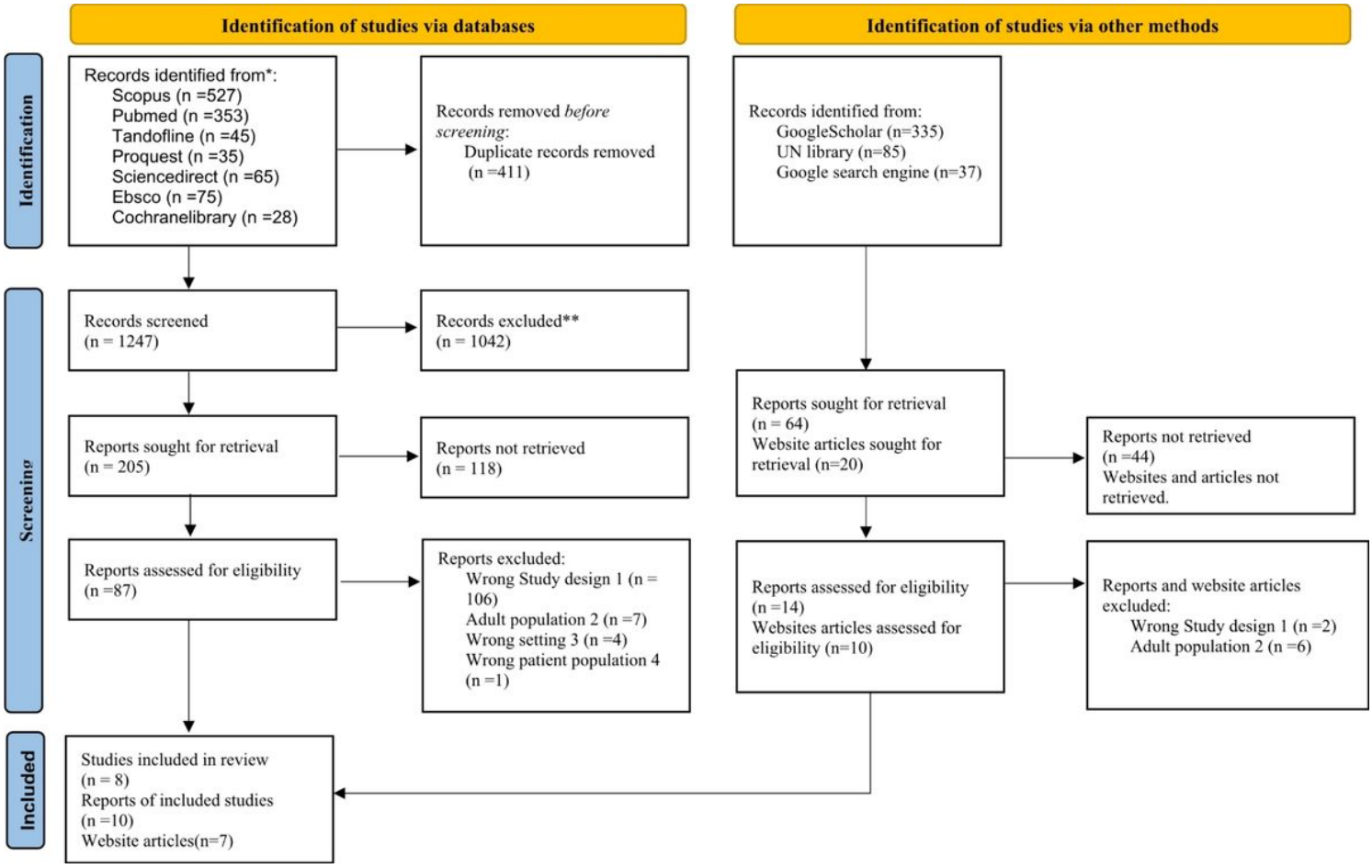

Figure 1

PRISMA flow diagram of the selected studies and reports

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